



Airoha IoT SDK for BT Data Firmware Update Developer's Guide

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Document Revision History

Revision	Date	Description
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1. Overview

Airoha IoT SDK for BT Data enables the Firmware update Over The Air (FOTA), a widely adopted cost and time efficient solution to update the firmware on the connected devices. The purpose of this developer's guide is to provide a complete description on how to deploy FOTA on the NOR flash.

This document guides you through:

- FOTA architecture layout
- FOTA update on the Airoha IoT development platform
- FOTA update workflow

1.1. Architecture Layout

Figure 1 shows the FOTA architecture for updating the firmware through Bluetooth using RACE CMD.

- The **FOTA Update Agent** is part of the **Bootloader** and uses the flash APIs to process the update on the target device.
- The **RACE CMD protocol** on the Airoha IoT device and the mobile device is used for exchanging data between the two devices through Bluetooth or USB. The **FOTA application** sends the FOTA package to the Airoha IoT device and triggers the Airoha IoT device to reboot and start the FOTA upgrade process.
- The **FOTA packaging tool** is used on the PC to apply the Lempel–Ziv–Markov chain algorithm (LZMA) compression and sign the signature in the new firmware file and generate a FOTA update package.

A description of the numbered items in Figure 1 is shown below:

- 1) **FOTA packaging** — Generate an updated package with or without FOTA package tool.
- 2) **HTTP server** — The storage location of the updated package.
- 3) **FOTA application** — The mobile device downloads the updated package from the server.
- 4) **RACE CMD protocol** — The **FOTA application** pushes the updated package to the Airoha IoT device using Bluetooth or USB.
- 5) **FOTA RACE CMD** — Sets the triggered flag and reboots the Airoha IoT device through **FOTA application** when the complete package is received.
- 6) **Update Agent** — Checks the flag status and proceeds with the update.

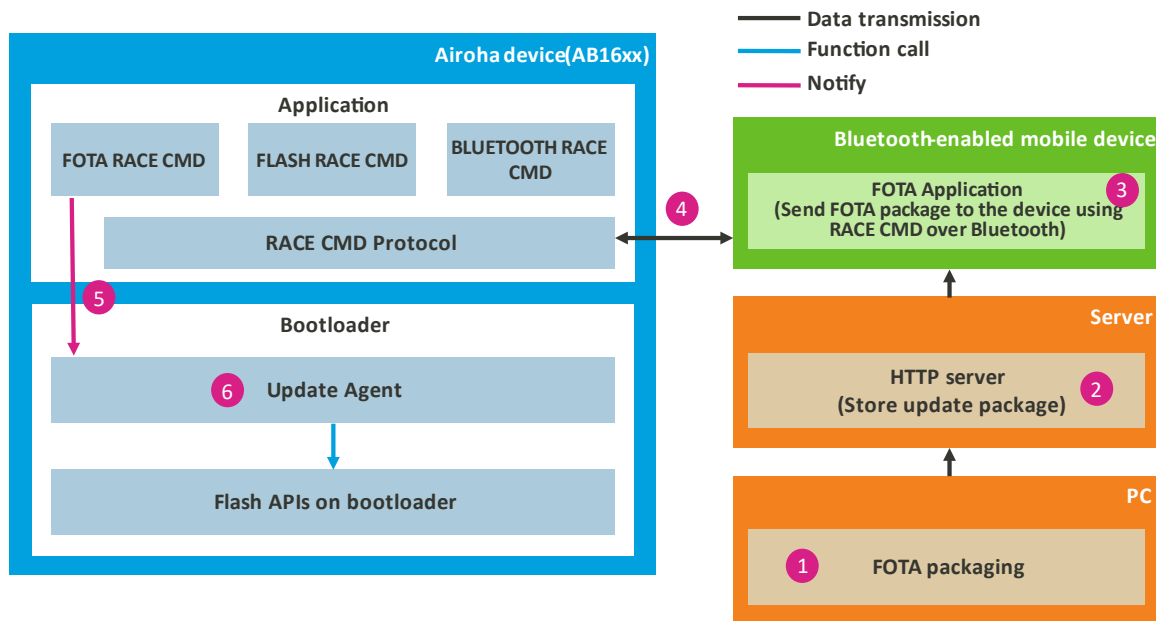


Figure 1. FOTA Architecture Layout for Updating the Firmware through Bluetooth using RACE CMD

1.2. Feature Support Status

The Airoha IoT SDK for BT Data development platform is an integrated solution with limited availability of FOTA features on each chipset. Table 1 shows the feature support summary on each chipset.

Table 1. FOTA Feature Support Status on Chipsets

Chipset	Device firmware upgrade	Update through Bluetooth
AB1620	✓	✓
AB1615	✓	✓

2. Using the FOTA Update

2.1. FOTA Packaging

There are two types of FOTA methods based on the packaging approach: Compressed FOTA and Uncompressed FOTA. Compressed FOTA requires the use of the FOTA package tool to generate a compressed updated package. In contrast, Uncompressed FOTA directly transfers the raw binary; thus, the raw binary serves as the updated package without needing the FOTA package tool. AB1615 only supports the Uncompressed FOTA.

2.1.1. Compressed FOTA Packaging

Airoha provides a FOTA packaging tool that runs on the Microsoft Windows OS to compress data, generate a checksum, and prefix a header for the new binary FOTA package file. During the update, the **Update Agent** parses the header, validates the package file using the checksum, and identifies where the data should be written. A FOTA package file is generated by using the FOTA package tool. Figure 2 shows the UI of the FOTA packaging tool.

FOTA Package Tool-1.0.0.213

Password: Login

Step 1. Load Partition info. from ...

☒ from flash_download.cfg

Step 2. Select partitions

Selection	Partition Name	SourceType	Offset	Length	Target Address	Source Filename
-----------	----------------	------------	--------	--------	----------------	-----------------

Step 3. Set options and generate FOTA package

☒ Enable LZMA

☒ Enable AES

AES Key: 0x

AES IV: 0x

Version:

☐ reserved nvdm item update

Generate FOTA package

Figure 2. UI of FOTA Packaging Tool

To use the FOTA packaging tool:

- 1) Double-click Airoha.Tool.Kit.exe.
- 2) Click the **"FOTA Package Tool"** button.
- 3) Click the **"from flash_download..."** button and select the specific flash_download.cfg file. The flash_download.cfg file is generated automatically after the build flow. It can be found under the sub folder of <sdk_root>/out. Normally, it contains the information of the flash partitions that need to be updated.

- 4) Select the Flash Partition(s) to upgrade.

When more than one bin is selected in the UI, the FOTA package file contains information for those bins. The device upgrades multiple bins in multiple Flash sectors during the execution of the FOTA upgrade process.

- 5) Select whether “**Enable LZMA**” compression is applied.
- 6) Select whether “**Enable AES**” encryption is applied.

AES depends on LZMA. The AES checkbox is only available when the LZMA checkbox is selected. The AES key and AES IV set in the tool must be the same as the `g_aes_key` and the `g_aes_iv` in the `<sdk_root>/airoha/mcu/applications/mcuboot/boot/fota/src/bl_fota_upgrade.c`.

(**Note:** mcuboot is the bootloader responsible for firmware upgrades.)

- 7) Click the “**Generate FOTA package**” button to create the final FOTA package.

2.1.2. Uncompressed FOTA Packaging

Uncompressed FOTA does not use the FOTA package tool to create the update package. Instead, it directly uses the raw binary that is produced after the build process as the update package.

The path for the Uncompressed FOTA updated package is:

```
<sdk_root>/out/ab16xx_xxx/<project_name>/<project_name>/zephyr/zephyr.signed.confirmed.bin
```

2.2. FOTA Update Methods

2.2.1. Compressed FOTA Update

This update method is easy to use and does not occupy much flash memory space if a compression algorithm is applied to reduce the size of the updated package.

A new FOTA package file is generated using the FOTA package tool. There is a reserved space on the NOR Flash to store the new FOTA package file. When the new FOTA package file is successfully transferred and stored in the specified address of the reserved space, the system reboots automatically. Normally, the system reboots to enter the bootloader phase.

When the bootloader detects that there is a new FOTA package file on the flash, it moves the new bin file(s) included in the FOTA package file into the corresponding flash partitions. After it is done successfully, the system reboots with the new bin file(s).

The compile options for the Compressed FOTA update are configured in the main project and bootloader project config file (feature.conf). One of the config files is in the main application folder, and the other is in the bootloader project folder that builds the bootloader binary. The FOTA feature is enabled only when the FOTA options in both are enabled.

Configure the following options in `PROJECT_DIRECTORY/config/ab16xx_xxx/feature.conf` of the main application.

<code>CONFIG_AIR_FOTA_ENABLE=y</code>
<code>CONFIG_AIR_DUAL_BANK_FOTA_ENABLE=n</code>

The same options also need to be configured for the bootloader's config file under `MCUBOOT_PROJECT_DIRECTORY/ boot/zephyr/config/ ab16xx_xxx/feature.conf` in the bootloader project folder, as shown below:

<code>CONFIG_AIR_FOTA_ENABLE=y</code>
<code>CONFIG_AIR_FOTA_UPGRADE_ENABLE=y</code>

CONFIG_BOOT_UPGRADE_ONLY=n

2.2.2. Uncompressed FOTA

The FOTA application will push the raw binary as an updated package to the device; then, it reboots the device to update the binary file.

The compile options for the dual device FOTA update are configured in the main project and the bootloader project config file (feature.conf). One of the config files is in the main application folder. The other config file is in the bootloader project folder that builds the bootloader binary. The dual device FOTA update feature is only enabled when the FOTA options in both config files are enabled.

Configure the following options in PROJECT_DIRECTORY/config/ab16xx_xxx/feature.conf of the main application.

CONFIG_AIR_FOTA_ENABLE=y
CONFIG_AIR_DUAL_BANK_FOTA_ENABLE=y

Configure the following options in MCUBOOT_PROJECT_DIRECTORY/ boot/zephyr/config/ ab16xx_xxx/feature.conf of the bootloader project.

CONFIG_AIR_FOTA_ENABLE=y
CONFIG_AIR_FOTA_UPGRADE_ENABLE=n
CONFIG_BOOT_UPGRADE_ONLY=y

2.3. Suggested Flash Layout Setting for Each Chip with FOTA Enabled

2.3.1. Compressed FOTA Buffer Partition Configuration

Figure 3 shows the default flash layout with Compressed FOTA enabled.

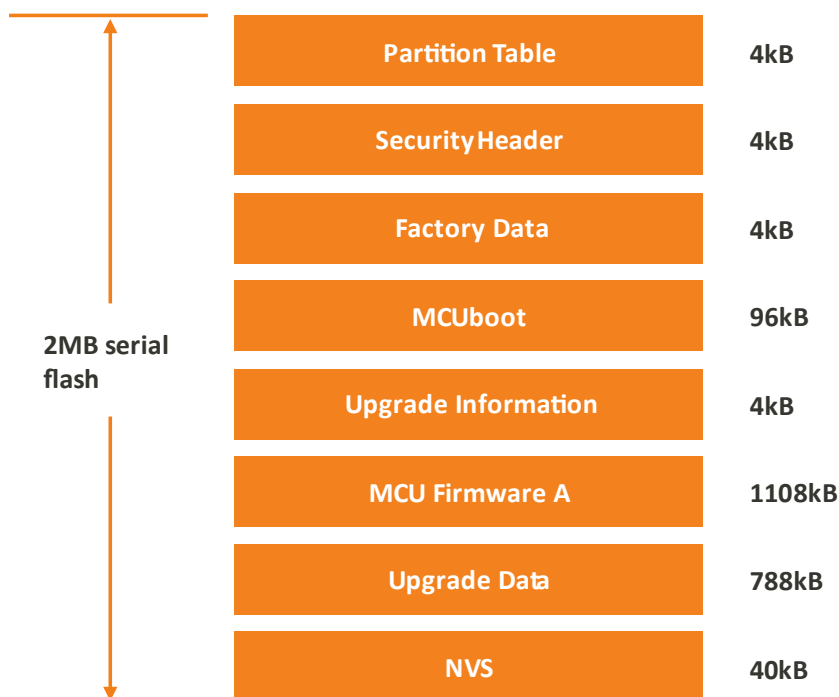


Figure 3. AB16xx Compressed FOTA Flash Layout Default Settings

The FOTA buffer partition information is read from the partition table when the FOTA package is stored in the internal Flash. To change the FOTA buffer partition size, changes must be made to the partition table.

- Refer to the Memory Layout Developer's Guide under <sdk_root>/airoha/mcu/doc for more information.
- Make any necessary changes to the FOTA buffer's start address and length.

2.3.2. Uncompressed FOTA Buffer Partition Configuration

Figure 4 shows the default flash layout with Uncompressed FOTA enabled.

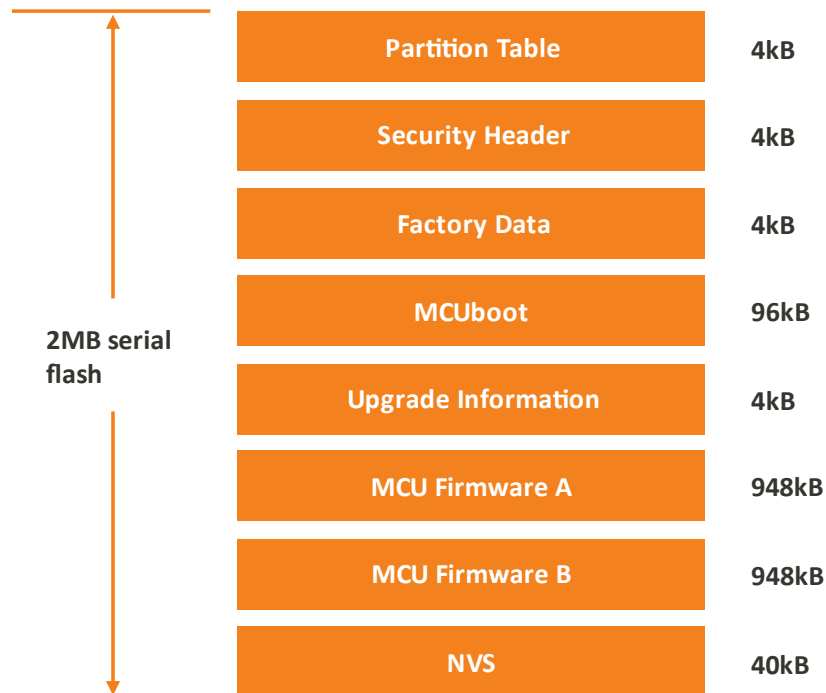


Figure 4. AB16xx Uncompressed FOTA Flash Layout Default Settings

3. FOTA Update Workflow

This section provides details on how to download the FOTA package and update the firmware in the bootloader.

FOTA resuming is supported during the FOTA download. When FOTA is stopped, the FOTA application records the current download position, and when FOTA is started again, it will continue sending the remaining FOTA package file to the device. In such a way, if the FOTA download did not finish last time, it can continue downloading the remaining part instead of always downloading the file from the very beginning.

FOTA cancel is also supported during the FOTA download. FOTA download can be cancelled from the FOTA application side or the Airoha IoT device side. When the FOTA download is cancelled, the FOTA application and the device terminate the FOTA download flow. To cancel FOTA download from the FOTA application side, press the **Cancel** button in the FOTA application. To cancel FOTA download from the Airoha IoT device side, call the `race_fota_cancel()` API. Even if the FOTA download is cancelled, the part of the FOTA package file that has already been downloaded is stored, allowing the FOTA download process to resume next time.

3.1. FOTA Package Download

When triggering the FOTA update process through the FOTA application, the FOTA application sends the FOTA package to the device through Bluetooth or USB using RACE CMD. First, the FOTA application sends RACE CMD to query the start address and the length of the FOTA buffer. It then sends RACE CMD to write the FOTA package to the flash directly. When the complete FOTA package is sent, it sends RACE CMD to request an integrity check of the FOTA package that was just received. If the integrity check passes, it sends RACE CMD to trigger a reboot of the device to execute an FOTA upgrade by the bootloader. Figure 5 shows the FOTA package download process.

The device remains passive during download and can only receive the RACE CMD from the FOTA application and execute the operations indicated by the RACE CMD.

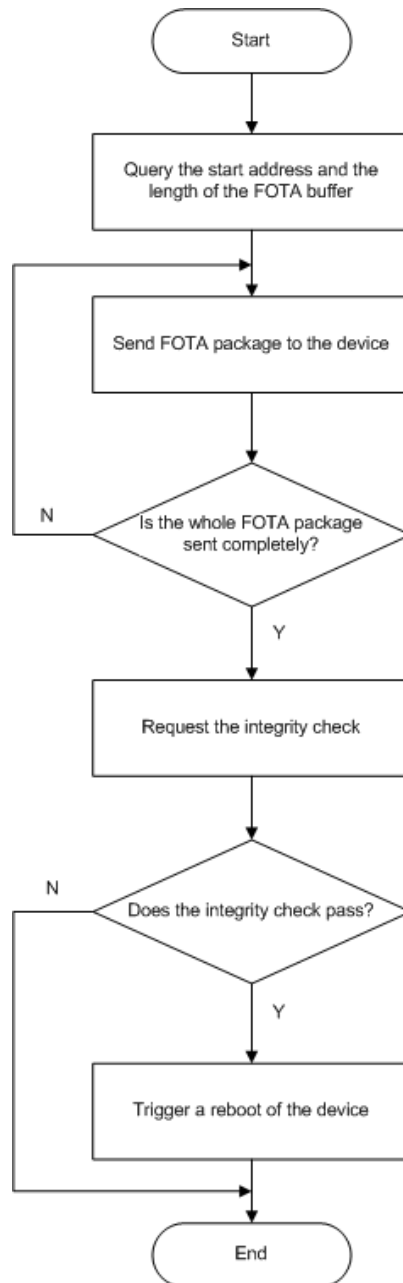


Figure 5. FOTA Package Download Workflow

3.2. FOTA Upgrade

3.2.1. Compressed FOTA Upgrade

During the reboot process, the bootloader checks if the last history record is 0x0100, which means that the FOTA upgrade process has been executed one time, but for some reason it is not complete. If the last history record is 0x0100, the bootloader starts moving the bin files included in the FOTA package from the FOTA buffer partition to the corresponding flash partitions. If the last history record is not 0x0100, it continues to check if the FOTA upgrade flag is set. If the flag is set, it also starts moving the bin files. If not, it goes into the normal boot up process. Figure 6 shows the FOTA upgrade workflow.

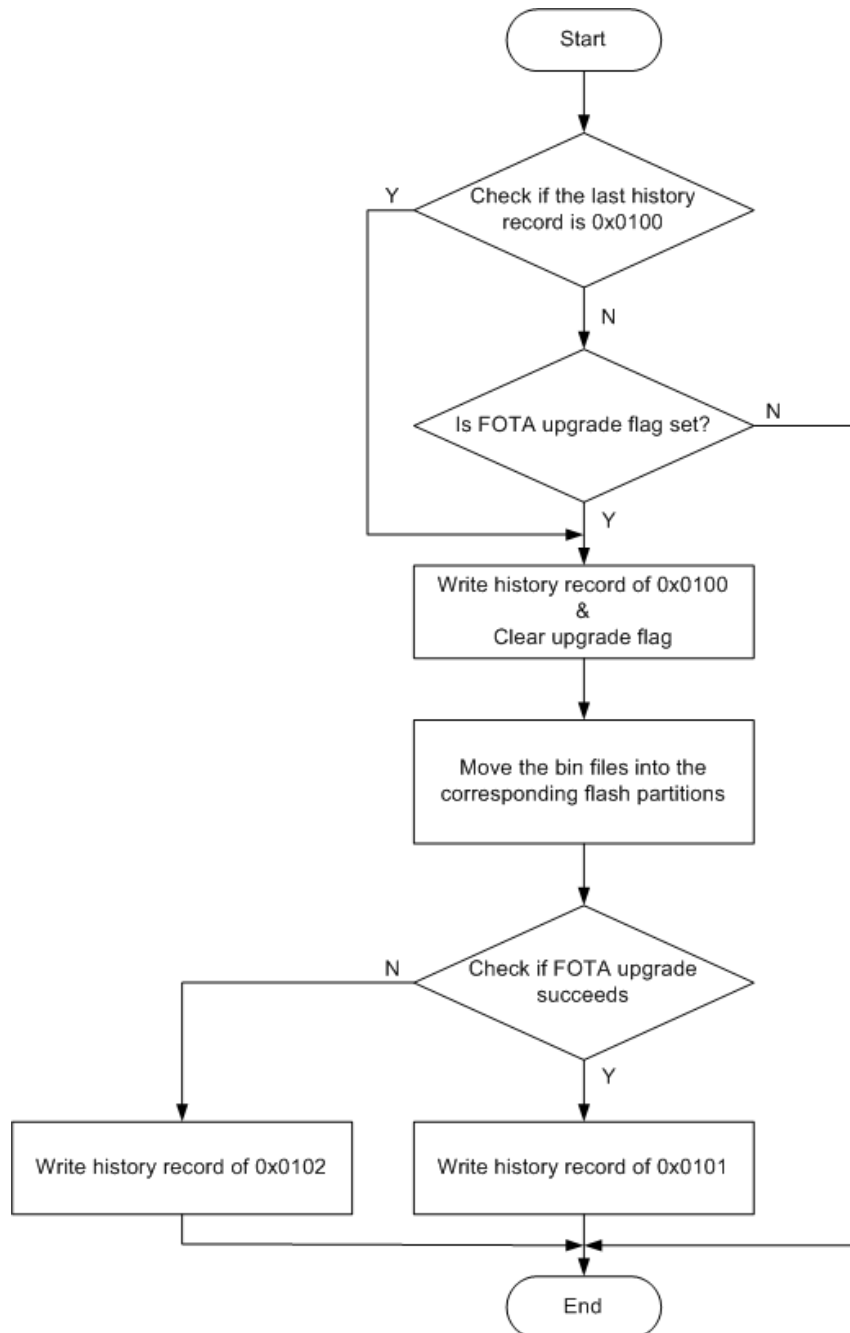


Figure 6. Compressed FOTA Upgrade Workflow

3.2.2. Uncompressed FOTA Upgrade

After downloading the updated package to MCU Firmware B, the system reboots into the bootloader. The bootloader then moves the raw binary from MCU Firmware B to MCU Firmware A and reboots from MCU Firmware A.

4. Appendix A: Acronyms and Abbreviations

Table 2 lists the acronyms and abbreviations used in this developer's guide.

Table 2. Acronyms and Abbreviations

Abbreviation / Term	Expansion / Definition
FOTA	Firmware over the air, a way to update firmware wirelessly.
NOR Flash	NOR Flash is a type of non-volatile storage that does not require power to retain data.

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